DATA ITEM DESCRIPTION

Title: MACHINERY AND EQUIPMENT PRE-INSTALLATION STRUCTUREBORNE SOUND MEASUREMENTS TEST REPORT

Number: DI-HFAC-80274A
AMSC Number: N9533
DTIC Applicable: No
Preparing Activity: SH
Applicable Forms:

Approval Date: 20150413
Limitation: N/A
GIDEP Applicable: No
Project Number: HFAC-2015-008

Use/relationship: The Machinery and Equipment Pre-Installation Structureborne Sound Measurements Test Report describes the performance and results of pre-installation structureborne sound tests for machinery and equipment.

a. This Data Item Description (DID) contains the format, contents, and preparation instructions for that data generated as specified in MIL-STD-1474 and ANSI/ASA S2.16, *Vibratory Noise Measurements and Acceptance Criteria of Shipboard Equipment*.

b. This DID shall be applied for each machinery and equipment item with a structureborne noise requirement, excluding fans, fan-coil units, fan-coil assemblies, air handling units, and other air moving devices.

c. The data reported by this DID may be used to support the analysis and design development described in DI-HFAC-81278, *Airborne Noise Analysis and Control Design History Booklet*.

d. This DID supersedes DI-HFAC-80274.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the contract.

2. Format. The Machinery and Equipment Pre-Installation Structureborne Sound Measurements Test Report shall be in the Contractor's format.

3. Content. The Machinery and Equipment Pre-Installation Structureborne Sound Measurements Test Report shall contain the following sections.

3.1. Test and equipment information. This section shall contain:

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a. Make, model, and serial number of the unit under test, including identification of major components, driver, and driven unit.
b. Ship name(s) and hull number(s) for which the unit is being acquired.
c. Compartment or on-deck location where the unit will be installed. If identical units by make and model number will be installed in any one compartment, identify the specific locations in that compartment where the unit will be installed.
d. Overall unit dimensions and orientation of machinery or equipment item undergoing testing.
e. Evaluating activity and test site, including name, telephone number, and e-mail addresses of the individuals who performed and supervised the tests.
f. Test date.
g. Specified acceptance criteria, including reference units.
h. Driver unit horsepower.
i. Rated electrical characteristics, including volts, amps, and phase.
j. Rated fluid media parameters (e.g., flow rates, temperatures, pressures).
k. Equipment rated operational speeds (e.g., driver and driven unit revolutions per minute (rpm)).
l. Expected equipment primary source excitation frequencies (e.g., rotational frequencies, gear tooth mesh frequencies, motor slot frequency, cylinder firing frequencies, vane frequencies, blade frequencies, piston frequencies, turbocharger frequencies).
m. Resilient mount manufacturer, make, model number, date of manufacture, load range, and natural frequency of resilient mounts used to support the device under test.
n. Number of resilient mounts used. A photo shall be included showing each installed resilient mount.
o. If the use of non-Navy standard resilient mounts has been approved, include the characteristics including load rating in pounds and resonant frequency of the mounts.
p. Operational load (e.g., full, half) of the machinery or equipment undergoing testing.
q. Operating speeds (rpm) used during test.
r. Throttling conditions, if any, and a description of how throttling conditions are set and verified.
s. Operating suction and discharge fluid media parameters during test (e.g., flow rates, pressures, temperatures).
t. Electrical inputs, outputs, or both used during test.
u. Identification of make and model of measurement and analysis instrumentation used, including calibration instrumentation. Certificates of laboratory calibration shall be included.
v. Details of instrument settings used during the testing, (e.g., type of sampling window, coupling, sampling frequency, sampling time, number of averages, averaging type, and frequency resolution of the data, if appropriate).

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w. Name and other information identifying any individuals representing the procuring activity who witnessed the test, including e-mail and telephone number.

3.2. Drawings, photographs, and calculations. This section shall contain:

a. Drawing and dimensional description of the subbase used to maintain alignment between the driver and driven unit.
b. Resilient mount location and orientation, including any sway braces; and overall wet weight and center of gravity of the item undergoing testing with respect to a defined reference position.
c. Calculated load on each resilient mount compared to specified load range of each mount; and calculated natural frequency of each mount when experiencing the calculated load.
d. Drawing and dimensional description of the foundation and/or mounting structure used to support the machinery or equipment item undergoing testing.
e. Calculations verifying compliance with the foundation natural frequency requirements of ANSI/ASA S2.16, and calculations verifying compliance with the mounting structure natural frequency and mass requirements of ANSI/ASA S2.16.
f. Calculations verifying that coincidence between the mounting system natural frequencies and the machine forcing frequencies do not exist.
g. Arrangement and description of flexible hose assemblies and flexible duct connections showing configuration, diameter and lengths of flexible sections, and details of how the weights of these were supported. Photos shall be included.
h. Arrangement of piping and fittings used to facilitate testing and details of how the weights of these were supported.
i. Arrangement of electrical power leads coupled with arrangements of cables used with instrumentation.
j. Position and orientation of each installed transducer (accelerometer) with respect to a defined reference position. A photograph shall be included showing where each transducer (accelerometer) is installed.
k. Description of any machinery items that were located nearby the test setup such as machinery located in an adjacent test cell, and the operational condition of those machinery items at the time period when the testing was undertaken.

3.3. Measured data. This section shall contain:

a. Measured data in reference units that are identical to the specified reference units.
b. Any measurements that are included in graphical form shall show ordinate (magnitude) values with 10 dB between major divisions.
c. Measured data compared, by measurement location and orientation, with accompanying information at that same location including background and ambient data and assigned limits.

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d. For structureborne noise levels that exceed the specified limits, excesses shall be clearly indicated in the tabulated and plotted data.
e. If an excess to the specified one-third octave band structureborne levels exist, plots of narrowband data measured at the offending transducer positions and orientations, with frequency resolution not to exceed 1 Hz, shall be included.
f. The report shall identify the mechanisms that exist within the machinery or equipment item that are specifically responsible for the measured excesses (e.g., rotational frequency, blade, vane or piston frequency, firing frequency, gear mesh frequency, coincidence of a forcing frequency with a natural frequency of the mounting system, resonance in the subbase).

End of DI-HFAC-80274A

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